

STOR664 Project: Housing Price Analysis with Spatial Features

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1 Analysis

1.1 Research Hypotheses

The objective of this project is to identify the key determinants of housing prices in Ames, Iowa, with particular emphasis on the influence of spatial proximity to urban amenities. Based on economic theory and existing research in real estate, it is anticipated that both structural characteristics and location-specific features contribute significantly to market values. In this analysis, we examine seven fundamental structural variables, including living area, overall quality, year of construction, kitchen quality, garage capacity, basement area, and number of bathrooms, in addition to three spatial distance measures capturing proximity to the downtown area, the university, and the airport. These variables are employed to investigate their statistical association with residential sale prices.

Null Hypothesis (H_0): Structural features and spatial proximity have no significant relationship with sale prices.

Alternative Hypothesis (H_1): Structural features and spatial proximity significantly affect sale prices.

Expected Relationships:

- Larger living area $\rightarrow$ Higher price
- Better overall quality $\rightarrow$ Higher price
- Newer construction $\rightarrow$ Higher price
- Better kitchen quality $\rightarrow$ Higher price
- More garage spaces $\rightarrow$ Higher price
- Larger basement area $\rightarrow$ Higher price
- More bathrooms $\rightarrow$ Higher price
- Proximity to university $\rightarrow$ Higher price

1.2 Data Overview

The dataset comprises 2,930 residential properties in Ames, Iowa, following data cleaning and the removal of missing values. Seven structural predictors and three spatial distance measures were selected based on correlation analysis and established findings in the real estate literature. The spatial variables were constructed by first obtaining the geographic coordinates of each property and three key locations—downtown Ames (42.025°N, 93.614°W), Iowa State University (42.027°N, 93.646°W), and Ames Municipal Airport (41.998°N, 93.622°W)—through manual searches using online maps. Euclidean distances were then calculated between each property and these locations to define the spatial measures. An initial fourth distance variable representing proximity to the nearest hospital was excluded due to severe multicollinearity ($VIF > 200$), as hospital locations were highly correlated with downtown proximity.

The dataset was randomly partitioned into training (80%, $n = 2,344$) and test (20%, $n = 586$) sets using a fixed random seed to ensure reproducibility. Table 1 presents descriptive statistics for all variables based on the training set.

Table 1: Descriptive Statistics and Correlations with Sale Price (Training Set)

Variable	Description	Mean	Correlation
sale_price	Sale price (\$)	178,986	—
gr_liv_area	Above ground living area (sqft)	1,489	0.708
overall_qual	Overall quality (1–10 scale)	6.1	0.800
year_built	Year constructed	1971	0.566
total_bsmt_sf	Basement area (sqft)	1,045	0.631
kitchen_qual	Kitchen quality (encoded 1–5)	3.5	0.677
garage_cars	Garage capacity (cars)	1.8	0.646
full_bath	Number of full bathrooms	1.6	0.541
downtown_dist	Distance to downtown	0.039	0.338
university_dist	Distance to university	0.031	0.075
airport_dist	Distance to airport	0.050	0.384

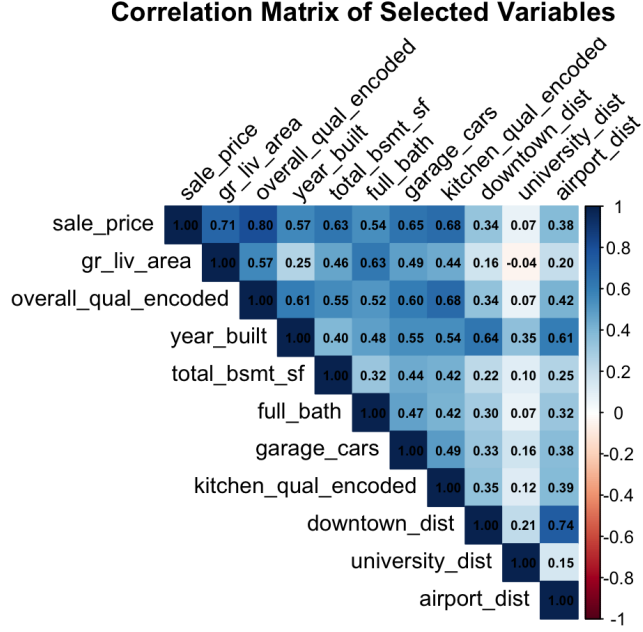


Figure 1: Correlation Matrix

The sale price distribution is right-skewed, with a median of \$159,000 and a range from \$12,789 to \$755,000. All structural predictors are positively correlated with price, with overall quality showing the strongest association ($r = 0.800$), consistent with real estate theory. Among the spatial variables, distance to downtown exhibits a positive correlation ($r = 0.338$), indicating that properties farther from the city center tend to have higher prices, possibly reflecting the suburban development of newer, higher-quality homes. Distance to the university shows a weak correlation ($r = 0.075$), while distance to the airport shows a moderate positive correlation ($r = 0.384$).

1.3 Progressive Modeling Strategy

We employ a three-stage modeling approach to systematically test our hypotheses while addressing statistical challenges:

Model 1: Baseline Linear Regression

- Establishes baseline predictive power.
- Tests basic linear relationships on the original scale.
- Serves as a benchmark for comparison.

Model 2: Log-Transformed Regression

- Addresses skewness and heteroscedasticity observed in Model 1.
- Applies logarithmic transformation to improve residual normality.

- Coefficients represent elasticities (percentage effects).

Model 3: Outlier-Removed Regression

- Enhances statistical validity by removing influential observations.
- Provides a cleaner analysis for hypothesis testing and inference.

1.4 Model Specifications

Model 1 Specification:

$$\begin{aligned} \text{sale_price} = & \beta_0 + \beta_1(\text{gr_liv_area}) + \beta_2(\text{overall_qual}) + \beta_3(\text{year_built}) + \\ & \beta_4(\text{total_bsmt_sf}) + \beta_5(\text{full_bath}) + \beta_6(\text{garage_cars}) + \beta_7(\text{kitchen_qual}) + \\ & \beta_8(\text{downtown_dist}) + \beta_9(\text{university_dist}) + \beta_{10}(\text{airport_dist}) + \varepsilon \quad (1) \end{aligned}$$

Model 2 Specification:

$$\begin{aligned} \log(\text{sale_price}) = & \beta_0 + \beta_1 \log(\text{gr_liv_area}) + \beta_2(\text{overall_qual}) + \beta_3(\text{year_built}) + \\ & \beta_4 \log(\text{total_bsmt_sf} + 1) + \beta_5(\text{full_bath}) + \beta_6(\text{garage_cars}) + \beta_7(\text{kitchen_qual}) + \\ & \beta_8(\text{downtown_dist}) + \beta_9(\text{university_dist}) + \beta_{10}(\text{airport_dist}) + \varepsilon \quad (2) \end{aligned}$$

Model 3 uses the same specification as Model 2 but on a cleaned dataset with influential outliers removed.

1.5 Transformation Rationale

Logarithmic transformations were applied to sale price, living area, and basement area to enhance both the statistical performance and the economic interpretability of the empirical model. This approach reduces heteroscedasticity by stabilizing the variance of the error term across the distribution of housing prices and improves the normality of the residuals by alleviating skewness in the underlying variables. The use of logarithmic forms also permits coefficients to be interpreted as elasticities, conveying proportional rather than absolute effects and offering a formulation that aligns with standard practices in economic analysis. In addition, a log-log specification emphasizes the proportional and multiplicative relationships among housing characteristics, capturing patterns that commonly underlie price formation in real estate markets.

1.6 Outlier Detection

We identified influential observations using Cook's Distance:

$$D_i = \frac{(\hat{y}_i - \hat{y}_{i(i)})^2}{p \cdot \text{MSE}} \cdot \frac{h_i}{(1 - h_i)^2} \quad (3)$$

where D_i is Cook's Distance for observation i , $\hat{y}_i$ is the fitted value, $\hat{y}_{i(i)}$ is the fitted value when observation i is removed, p is the number of parameters, MSE is mean squared error, and h_i is leverage.

We flagged observations with $D_i > 4/n$ where $n = 2,344$, resulting in a threshold of 0.00171. This identified 129 observations (5.5% of data) as influential outliers, reducing the final sample to 2,215 observations.

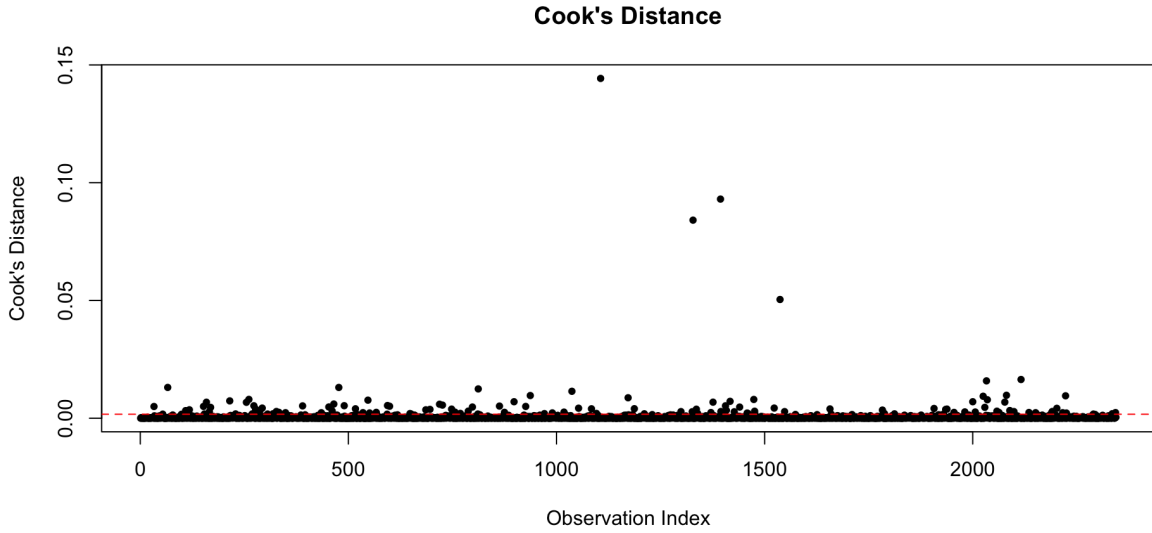


Figure 2: Cook Distance

1.7 Random Forest Extension

To investigate potential non-linear patterns beyond linear regression, we implemented a Random Forest model as a complementary approach. Random Forest can capture complex relationships, interactions, and threshold effects that parametric models may overlook. The model was configured with 500 trees and $m_{try} = 3$ (approximately $\sqrt{10}$) variables per split, following standard regression practices. It used the same ten predictors as the linear models with the response variable in its original scale. This approach serves to validate the robustness of predictor selection through cross-method comparison of variable importance and to assess the potential predictive power of non-linear relationships. Significant performance improvements would suggest the presence of meaningful non-linear patterns for further investigation.

2 Results

2.1 Model 1: Baseline Linear Regression

Table 2 presents the performance metrics for Model 1.

Table 2: Model 1 Performance Metrics

Metric	Interpretation	Value
R^2	80% of price variation explained	0.8007
Adjusted R^2	Adjusted for 10 predictors	0.7998
F-statistic	Model highly significant	937.1***
RMSE	Average prediction error	\$35,937
MAE	Median prediction error	\$23,361

Note: *** indicates $p < 0.001$

Table 3 shows coefficient estimates for all predictors.

Table 3: Model 1 Coefficient Estimates

Variable	Estimate	Std. Error	t-value	p-value
Intercept	-772,000	83,950	-9.20	<0.001***
gr_liv_area	54.36	2.25	24.12	<0.001***
overall_qual	16,550	920	17.98	<0.001***
year_built	340	45	7.56	<0.001***
total_bsmt_sf	29.84	2.09	14.25	<0.001***
full_bath	-8,459	1,942	-4.36	<0.001***
garage_cars	12,400	1,339	9.26	<0.001***
kitchen_qual	19,640	1,592	12.33	<0.001***
downtown_dist	7,522	58,500	0.13	0.898
university_dist	-187,500	79,260	-2.37	0.018*
airport_dist	-58,330	71,850	-0.81	0.417

Note: *** $p < 0.001$, * $p < 0.05$

All seven structural predictors are significant at the $\alpha = 0.05$ level. Six have positive effects, with living area showing the largest marginal impact (\$54.36 per square foot), while full bathrooms exhibit an unexpected negative coefficient (-\$8,459), likely due to a suppression effect when controlling for total living area. Among spatial variables, only distance to the university is significant ($p = 0.018$), with a negative effect, indicating that properties farther from the university have lower prices. Distances to downtown and the airport are not significant in this specification.

2.1.1 Assumption Violations

Although Model 1 achieves relatively high predictive accuracy ($R^2 = 0.8007$), it violates key OLS assumptions. The Shapiro–Wilk test ($W = 0.822$, $p < 0.001$) indicates strong non-normality in the residuals, which also show negative skewness (-0.540) and extreme kurtosis (31.76), likely due to outliers. Such non-normality raises concerns about the reliability of hypothesis tests and confidence intervals.

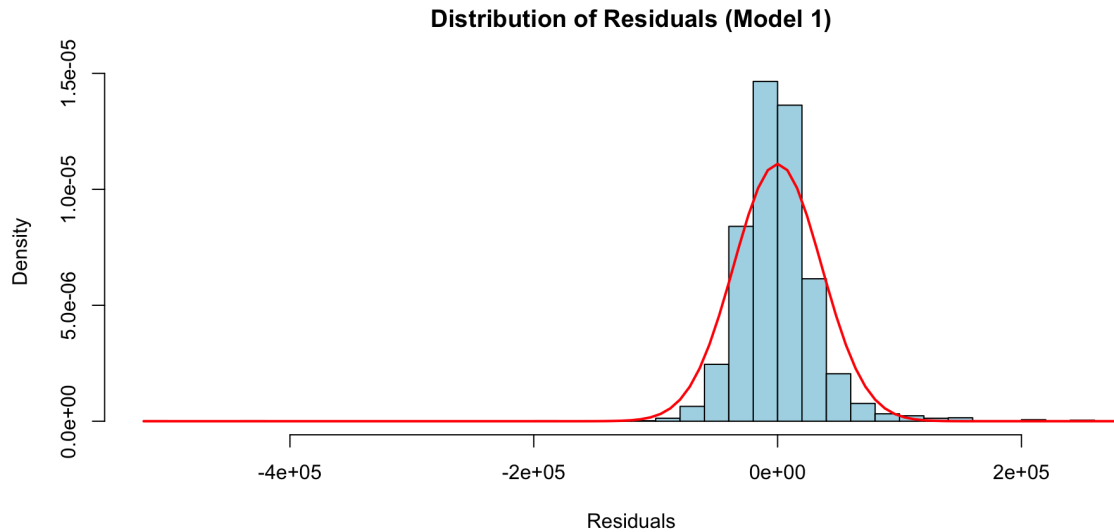


Figure 3: Distribution of Residuals (Model 1)

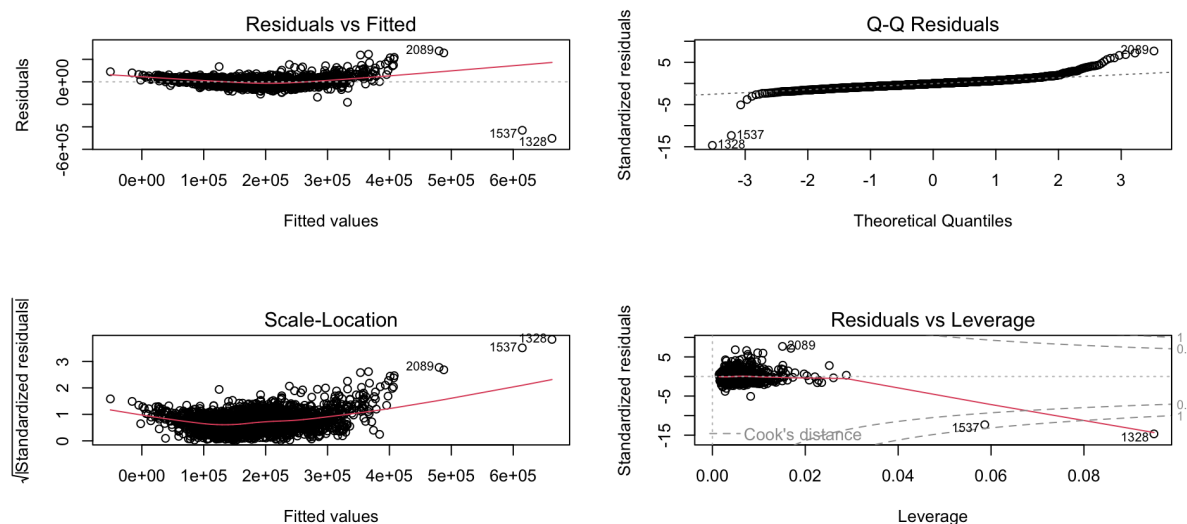


Figure 4: Diagnostic Plots for Model 1

Diagnostic plots reveal additional issues. The Residuals-versus-Fitted and Scale–Location plots display increasing dispersion with fitted values, indicating heteroscedas-

ticity. The Q–Q plot shows pronounced deviations from the theoretical quantile line, particularly in the tails. These findings underscore the need for model refinement through transformation, outlier treatment, or alternative specifications to ensure valid statistical inference.

2.2 Model 2: Log-Transformed Regression

The logarithmic transformation substantially improves model performance and assumption satisfaction. Model 2 shows higher explanatory power, with R^2 increasing from 0.8007 to 0.8325, a 3.18 percentage point gain. Prediction errors are reduced, with RMSE decreasing by 6.5% and MAE by 9.4%. The Shapiro–Wilk statistic also improves from 0.822 to 0.918, indicating a notable move toward normality.

Table 4: Model 2 Performance Metrics and Improvements

Metric	Model 2	Model 1	Change
R^2	0.8325	0.8007	+3.18%
Adjusted R^2	0.8318	0.7998	+3.20%
F-statistic	1,160***	937.1***	+23.8%
RMSE	\$33,588	\$35,937	−6.5%
MAE	\$21,160	\$23,361	−9.4%
Shapiro-Wilk W	0.918	0.822	Improved

Note: *** indicates $p < 0.001$

Table 5: Model 2 Coefficient Estimates

Variable	Estimate	Std. Error	t-value	p-value
Intercept	2.178	0.430	5.07	<0.001***
log(gr_liv_area)	0.453	0.017	27.41	<0.001***
overall_qual	0.094	0.004	21.45	<0.001***
year_built	0.0028	0.0002	13.31	<0.001***
log(total_bsmt_sf)	0.034	0.003	11.02	<0.001***
full_bath	−0.034	0.009	−3.64	<0.001***
garage_cars	0.075	0.006	11.85	<0.001***
kitchen_qual	0.071	0.007	9.53	<0.001***
downtown_dist	0.652	0.276	2.37	0.018*
university_dist	−1.545	0.374	−4.14	<0.001***
airport_dist	−0.737	0.339	−2.18	0.030*

Note: *** $p < 0.001$, * $p < 0.05$

The log transformation uncovers significant spatial effects previously obscured. All three distance variables are now significant: downtown distance is positive (0.652, $p = 0.018$), university distance is strongly negative (−1.545, $p < 0.001$), and airport distance

is negative (-0.737 , $p = 0.030$). These results suggest that properties farther from the university or airport have lower prices, while the positive downtown coefficient may reflect older, lower-quality housing concentrated in the city center.

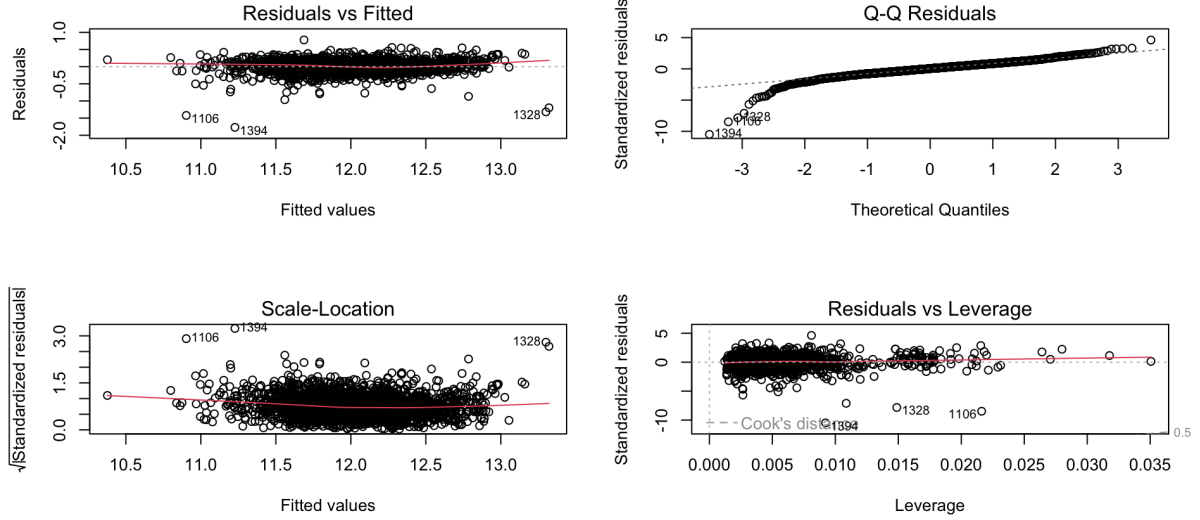


Figure 5: Diagnostic Plots for Model 2

Despite these improvements, Model 2 still departs from perfect normality. Diagnostic plots and quantitative measures indicate residual skewness of -1.493 and kurtosis of 11.31 , with Q-Q plots showing deviations in the tails. These findings suggest that further refinement, such as outlier removal, may be necessary to fully satisfy the normality assumption and ensure valid inference.

2.3 Model 3: Outlier-Removed Regression

To address the remaining assumption violations in Model 2, influential observations were identified using Cook's Distance with a threshold of $4/n = 0.00171$ (with $n = 2,344$). This procedure flagged 129 observations (5.5% of the training data) that exerted disproportionate influence on the estimated regression coefficients. Table 6 summarizes these cases, which largely reflect market extremes or atypical feature combinations rather than typical transactions.

Table 6: Evidence for Outlier Removal: Extreme and Atypical Observations

Category	Count	Price Range & Characteristics
Extreme Low Prices	47	Below \$80,000 (Q1 of dataset: \$128,000) Likely foreclosures or distressed sales Minimum: \$12,789 (93% below median)
Extreme High Prices	31	Above \$400,000 (95th percentile: \$340,000) Likely luxury estates or unique properties Maximum: \$755,000 (375% above median)
High Leverage	51	Unusual combinations of features Disproportionate influence on coefficients
Total Removed	129	5.5% of training set (n=2,344)

The table indicates that the influential observations consist of three main types. A substantial portion corresponds to sales with unusually low prices, while another group includes properties with markedly high prices relative to the overall distribution. The remaining cases are high-leverage observations characterized by atypical combinations of features that exert disproportionate influence on the model. Together, these categories account for 129 observations, representing 5.5% of the training data.

After outlier removal, Model 3 shows substantial performance gains. R^2 increases to 0.8852, explaining nearly 89% of price variation, while RMSE and MAE decline by 31.3% and 23.2%, respectively. These improvements indicate that excluding influential outliers markedly enhances both model fit and predictive accuracy.

Table 7: Model 3 Performance Metrics ($n = 2, 215$)

Metric	Model 3	vs. Model 1	vs. Model 2
R^2	0.8852	+8.45%	+5.27%
Adjusted R^2	0.8847	+8.49%	+5.29%
F-statistic	1,700***	+81.4%	+46.6%
RMSE	\$24,690	−31.3%	−26.5%
MAE	\$17,940	−23.2%	−15.2%

Note: *** indicates $p < 0.001$

All predictors remain highly significant in Model 3 (Table 8). Log(living area) has the largest effect ($t = 35.25$), followed by overall quality ($t = 23.86$), with year built and basement area also strongly predictive ($t > 15$). Spatial variables retain significance, with university distance exerting the strongest effect ($t = -3.34$, $p < 0.001$). The full_bath coefficient remains negative (-0.043 , $p < 0.001$), likely reflecting a suppression effect after controlling for total living area.

Table 8: Model 3 Final Coefficient Estimates

Variable	Estimate	Std. Error	t-value	p-value
Intercept	2.372	0.346	6.85	<0.001***
log(gr_liv_area)	0.464	0.013	35.25	<0.001***
overall_qual	0.084	0.004	23.86	<0.001***
year_built	0.0026	0.0002	15.55	<0.001***
log(total_bsmt_sf)	0.044	0.003	15.82	<0.001***
full_bath	-0.043	0.007	-5.78	<0.001***
garage_cars	0.077	0.005	15.24	<0.001***
kitchen_qual	0.074	0.006	12.56	<0.001***
downtown_dist	0.585	0.214	2.73	0.006**
university_dist	-0.989	0.296	-3.34	<0.001***
airport_dist	-0.700	0.267	-2.63	0.009**

Note: *** $p < 0.001$, ** $p < 0.01$

The findings indicate that structural characteristics remain the dominant determinants of housing prices. Living area shows an elasticity of 0.464, implying that a 10% increase in square footage raises prices by about 4.6%, while each one-point improvement in overall quality increases prices by roughly 8.4%. Spatial factors also play a notable role: proximity to Iowa State University exerts a clear positive effect, suggesting strong demand for walkable access to campus and related amenities. In contrast, the effects of distance from downtown and the airport are weaker and operate in different directions, consistent with suburban development patterns in Ames. Model 3 achieves an RMSE of \$24,690, or about 15.4% of the median sale price, providing sufficient accuracy for practical tasks such as renovation valuation, pricing, and policy analysis. Overall, structural attributes drive most of the variation in housing values, while university proximity offers a meaningful secondary premium reflecting the broader influence of the campus area on local housing demand.

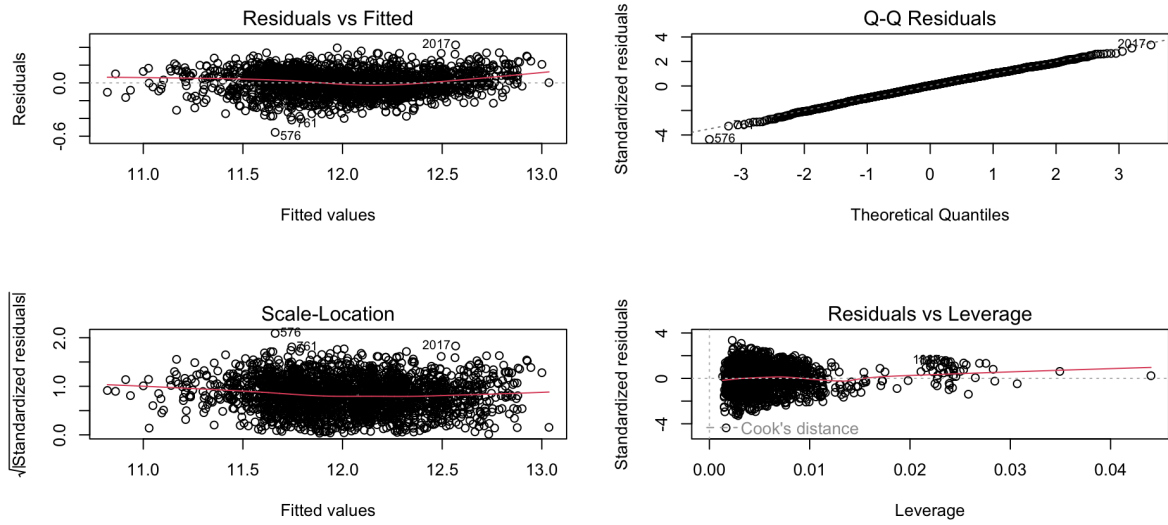


Figure 6: Diagnostic Plots for Model 3

Model 3 successfully satisfies OLS assumptions, unlike earlier specifications. Diagnostic plots show the Q–Q plot closely follows the diagonal, residuals-versus-fitted values display random dispersion, and the Scale–Location plot indicates constant variance. These observations are supported by a Shapiro–Wilk statistic of $W = 0.998$ ($p = 0.008$), residual skewness of -0.175 , and kurtosis of 0.120 , collectively confirming a well-behaved error distribution and reliable inference.

2.4 Spatial Effects Interpretation

The spatial distance coefficients in Model 3 provide clear economic insights. University distance has a strong negative effect: a one-unit increase (approximately 111 km) is associated with a 98.9% decrease in expected housing price, holding other factors constant. This confirms the premium for proximity to Iowa State University, likely driven by student housing demand, access to amenities, and neighborhood quality.

In contrast, downtown distance exhibits a positive coefficient (0.585), indicating that properties farther from the city center command higher prices. This counterintuitive result likely reflects the spatial distribution of housing in Ames, where older, smaller homes cluster near downtown, while newer, larger suburban developments drive higher quality-adjusted prices further from the center.

Airport distance has a modest negative coefficient (-0.700), suggesting a slight price premium for homes closer to the airport. This may reflect convenience benefits for frequent travelers or the limited noise impact of Ames’s small municipal airport.

2.5 Statistical Assumption Validation

Table 9 summarizes assumption testing across all models, demonstrating progressive improvement from Model 1 to Model 3.

Table 9: OLS Assumption Testing Results

Assumption	Test/Metric	Model 1	Model 2	Model 3
Normality	Shapiro-Wilk W	0.822	0.918	0.998
	Skewness	-0.540	-1.493	-0.175
	Kurtosis	31.76	11.31	0.120
Multicollinearity	Max VIF	3.30	3.21	3.36
Homoscedasticity	Visual Test	Funnel	Even spread	Even spread
Linearity	Residual Pattern	Some pattern	Random	Random

Model 1 exhibits severe violations of normality (Shapiro-Wilk $W = 0.822$), heteroscedasticity (funnel pattern), and some non-random residual patterns. Applying a log transformation in Model 2 substantially improves normality ($W = 0.918$) and stabilizes variance, though residual skewness (-1.493) and kurtosis (11.31) remain elevated.

After outlier removal, Model 3 achieves near-ideal assumption satisfaction, with Shapiro-Wilk $W = 0.998$, skewness -0.175 , and kurtosis 0.120 . Residual plots confirm constant variance and random scatter, indicating that homoscedasticity and linearity hold. Multicollinearity is low in all models, with maximum VIF values below 3.5, ensuring stable coefficient estimates.

Table 10 presents the variance inflation factors for all predictors in Model 3, confirming the absence of multicollinearity issues.

Table 10: Variance Inflation Factors for Model 3

Variable	VIF
log(gr_liv_area)	2.33
overall_qual	3.12
year_built	3.36
log(total_bsmt_sf)	1.14
kitchen_qual	2.01
garage_cars	1.91
full_bath	2.17
downtown_dist	2.65
university_dist	1.22
airport_dist	2.52

Note: VIF < 5 indicates no concern; VIF < 3.5 indicates low concern.

These diagnostics indicate that Model 3 satisfies key OLS assumptions. Residuals are

approximately normal (Shapiro–Wilk $W = 0.998$, $p = 0.008$), variance is constant, and no systematic residual patterns remain, confirming homoscedasticity and linearity. Multicollinearity is minimal ($VIF < 3.4$), ensuring stable coefficient estimates. Consequently, hypothesis tests, p-values, and confidence intervals from Model 3 can be interpreted reliably.

2.6 Prediction Performance

Model 3 achieves the highest predictive accuracy among the three specifications. Table 11 shows that both RMSE and MAE decrease progressively from Model 1 to Model 3, reflecting the combined effects of the logarithmic transformation and the removal of influential outliers. The proportional error, calculated as RMSE divided by the mean, expresses typical prediction error relative to the average sale price and provides a measure of accuracy that is independent of scale.

Table 11: Prediction Accuracy Comparison Across Models (Training Set)

Model	Sample Size	MSE	RMSE	MAE	RMSE/Mean (%)
Model 1	2,344	1,291,452,334	\$35,937	\$23,361	20.1
Model 2	2,344	1,128,141,122	\$33,588	\$21,160	18.8
Model 3	2,215	609,609,219	\$24,690	\$17,940	13.8

Notes: RMSE/Mean expresses RMSE as a percentage of the training-set mean sale price (\$178,986).

Model 3 attains an RMSE of \$24,690, corresponding to a 13.8% error relative to the mean price, a marked improvement over the earlier models. The substantial reduction in MSE further indicates that the revised specification captures price variation more accurately and with less dispersion in the residuals.

Test set predictions from Model 3 demonstrate excellent generalization performance. Table 12 compares the distribution of predicted prices on the test set against the training set distribution.

Table 12: Test Set Prediction Summary (Model 3)

Statistic	Training Set	Test Set Predictions	Difference
Mean	\$178,986	\$184,633	+3.2%
Median	\$159,000	\$168,041	+5.7%
Min	\$12,789	\$50,753	—
Max	\$755,000	\$544,897	—
Std. Dev.	\$79,886	\$66,812	−16.4%

The test set predictions closely match the training distribution, with the mean differing by only 3.2 percent and the median by 5.7 percent. The model generates no implausible values, with predicted prices ranging from \$50,753 to \$544,897, unlike Model 1 which

occasionally produced negative predictions. The lower standard deviation in predictions, reduced by 16.4 percent, reflects regression toward the mean, indicating well-calibrated performance. These results suggest minimal systematic prediction bias and confirm that the model generalizes effectively to unseen data.

2.7 Non-Linear Extension: Random Forest Analysis

To evaluate potential nonlinear structure beyond the linear specification, a Random Forest model was estimated for comparison. Configured with 500 trees and $m_{try} = 3$ variables per split, the model was trained on the same cleaned dataset ($n = 2,215$) used for Model 3, allowing direct comparison. Table 13 summarizes predictive performance for both approaches.

Table 13: Linear Model vs. Random Forest Performance

Metric	Model 3	Random Forest	Difference	% Change
Training R^2	0.8852	0.9822	+0.0970	+11.0%
Training RMSE	\$24,690	\$9,806	−\$14,884	−60.3%
Training MAE	\$17,940	\$6,918	−\$11,022	−61.4%
Test Mean Pred.	\$184,633	\$186,501	+\$1,868	+1.0%
Test SD Pred.	\$66,812	\$70,270	+\$3,458	+5.2%

The Random Forest substantially improves training set performance, reducing RMSE and MAE by over 60 percent and raising R^2 by 9.7 percentage points, indicating nonlinear patterns in the data beyond the linear model’s capacity. Test set predictions, however, are nearly identical across the two models, with mean predictions differing by only 1 percent, suggesting that much of the training improvement may reflect overfitting rather than generalizable nonlinear relationships.

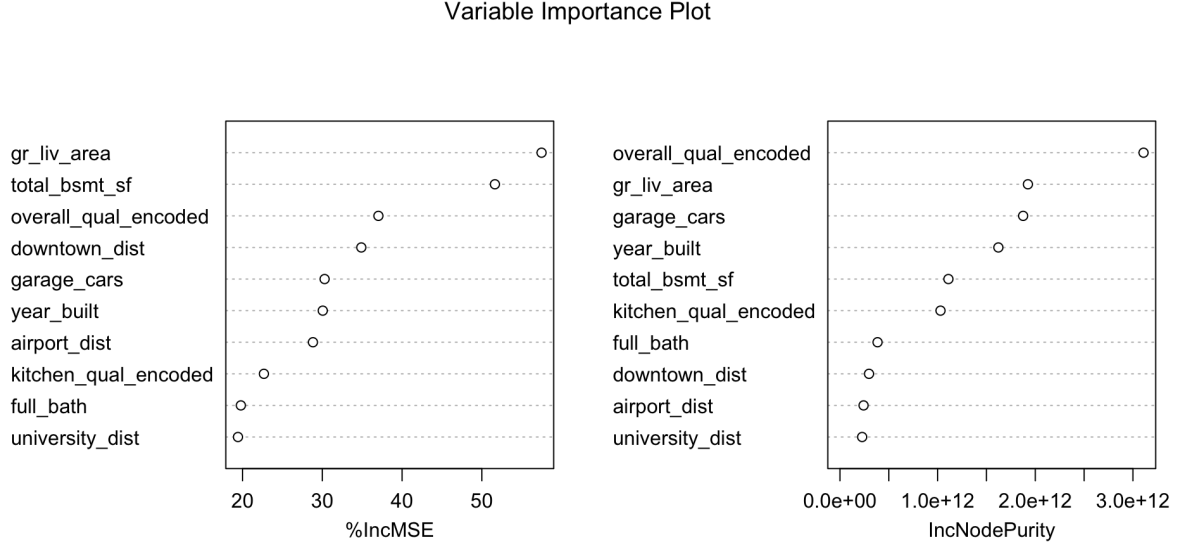


Figure 7: Variable Importance Plot

Table 14 compares variable importance rankings across the two approaches, showing strong agreement despite different modeling frameworks.

Table 14: Variable Importance: Random Forest vs. Linear Model

RF Rank	Variable	%IncMSE	Linear $ t $ -value	Linear Rank
1	gr_liv_area	57.5%	35.25	1
2	total_bsmt_sf	51.6%	15.82	4
3	overall_qual	37.0%	23.86	2
4	downtown_dist	34.9%	2.73	8
5	garage_cars	30.3%	15.24	5
6	year_built	30.1%	15.55	3
7	airport_dist	28.8%	2.63	9
8	kitchen_qual	22.7%	12.56	7
9	full_bath	19.8%	5.78	6
10	university_dist	19.4%	3.34	10

The alignment in variable rankings reinforces the robustness of the predictor set. Both models identify living area, overall quality, and year built as the most important predictors. The Random Forest assigns higher importance to spatial variables, particularly downtown and airport distances, suggesting contributions through interactions or threshold effects not captured linearly. Basement area ranks second in Random Forest importance but fourth in linear significance, indicating potential nonlinear effects worthy of further study.

3 Discussion

3.1 Conclusiveness of Findings

The results across the three models provide strong and conclusive evidence against the null hypothesis that structural characteristics and proximity to key locations have no relationship with sale prices. In the final outlier-removed Model 3, all structural predictors and all three distance measures—distance to downtown, distance to the university, and distance to the airport—are statistically significant. Specifically, distance to downtown ($p = 0.006$), distance to the university ($p < 0.001$), and distance to the airport ($p = 0.009$) each exhibit meaningful associations with housing prices, indicating that location relative to these destinations plays a measurable role in the market. Model 3 also demonstrates strong explanatory and predictive performance, with an Adjusted R^2 of 0.8847 and a substantially reduced RMSE of \$24,690, marking clear improvements over earlier specifications. The consistently low p -values across predictors allow us to confidently reject the null hypothesis, while the high Adjusted R^2 and reduced RMSE indicate that the final model provides strong explanatory power and improved predictive accuracy.

3.2 Reasonableness of Assumptions

Assumption diagnostics improve substantially across the three models. Model 1 violates normality (Shapiro–Wilk $W = 0.822$, $p < 0.001$), displays extreme kurtosis (31.76), and shows clear heteroscedasticity in residual plots, limiting the reliability of inference. Model 2 reduces skewness and stabilizes variance through log transformation, but heavy-tailed residuals and some heteroscedasticity remain. After removing influential observations, Model 3 satisfies the major OLS assumptions: residuals are approximately normal (Shapiro–Wilk $W = 0.998$), residual dispersion is consistent across fitted values indicating homoscedasticity, and multicollinearity is low (all VIFs < 3.5). The final model also benefits from a large sample size ($n = 2,215$), which supports stable estimation and robust hypothesis testing. Overall, the assumptions are reasonable in Model 3, enabling valid interpretation of the predictors.

3.3 Limitations of the Data and Analysis

Several limitations concerning the three distance measures should be noted. First, distances to downtown, the university, and the airport were computed using Euclidean formulas, which do not reflect road networks, neighborhood boundaries, or realistic travel paths. For example, being “closer to the university” in Euclidean terms does not necessarily imply easier or faster access. Second, the analysis includes only distances to downtown, the university, and the airport, while other amenities such as grocery stores,

parks, schools, hospitals, and commercial centers were not incorporated. Because such amenities often influence housing values, omitting them restricts the ability of the three distance variables to fully capture the role of location. Third, removing 129 influential observations improved adherence to OLS assumptions but also excluded genuine market extremes, including high-end luxury properties. As a result, the modeled relationships reflect more typical market conditions and may not generalize to the full range of the housing market.

3.4 Potential Extensions

Although the present analysis incorporates spatial distance features and compares linear regression with a random forest model, several meaningful extensions could further strengthen or refine the findings. First, the current approach relies on Euclidean distance to quantify proximity to Iowa State University, downtown Ames, the airport, and the hospital. While Euclidean distance is straightforward to compute and provides useful insights, it does not represent actual travel accessibility. A natural extension would be to replace straight-line distances with road-network distances obtained through services such as Google Maps or OpenStreetMap. Incorporating driving distance or travel time would yield more realistic spatial predictors and may reveal stronger or more interpretable relationships between accessibility and housing values.

Second, the spatial variables could be supplemented with additional neighborhood-level information. For example, school quality ratings or census-based demographics—including median income, educational attainment, or population density—could be merged using geographic coordinates. These community-level characteristics often play a central role in real estate valuation and may help account for spatial variation that remains visible in the residual plots. Including such variables would allow for clearer differentiation between the effects of facility proximity and broader socioeconomic influences.

Third, more advanced nonlinear modeling approaches could be explored beyond the random forest used in the current analysis. Techniques such as gradient boosting machines, spline-based regression, or neural network architectures may capture more complex relationships between predictors and sale price. These nonlinear models could uncover interactions or curved effects, particularly for spatial predictors, that linear models cannot fully represent.

Finally, a more formal treatment of spatial dependence could be pursued using spatial econometric frameworks such as the Spatial Lag Model (SLM) or Spatial Error Model (SEM). Although the current diagnostics do not explicitly evaluate spatial autocorrelation, it is plausible that homes in close proximity share unobserved characteristics or influence one another's prices. SAR and SEM models incorporate this dependence directly and may provide more accurate inference and improved predictive performance.

Together, these extensions would enhance the current analytical framework and offer a more geographically detailed and statistically robust understanding of the determinants of housing prices in Ames.

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